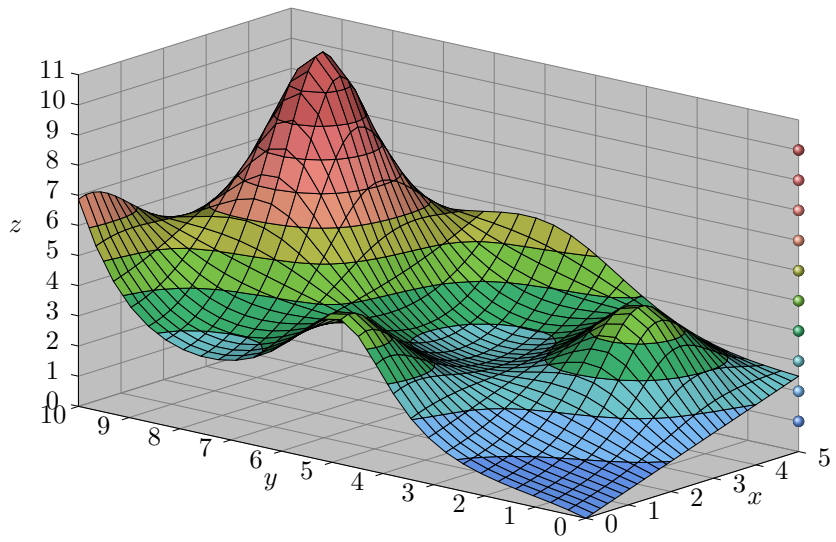
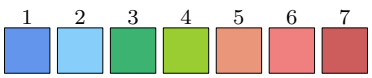
I. Motivations

This project was born out of a desire to enhance `luadraw` with a set of color palettes to easily produce graphics like the 3D plot shown below. Despite the availability of palette generation tools such as `Coolors`, the choice has been made to offer pre-built palettes, because this serves several key purposes for non-designers.

- **Reduced cognitive load:** pre-configuration eliminates design decisions, allowing to focus on content.
- **Consistency:** it becomes easy to ensure visual coherence across documents and projects, regardless of technology.
- **Accessibility compliance:** some pre-configured palettes guarantee colorblind-safe combinations.
- **Perceptual uniformity:** most palettes ensure distinguishable and meaningful colors for data visualization, while others serve artistic purposes.



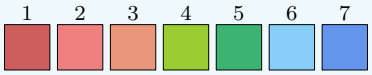
Technically, to generate the graphics in this section, `luadraw` is provided with a finite list of colors and uses linear interpolations to calculate the intermediate color values. In the example above, the finite color palette used is defined as follows.



Using this palette, `luadraw` can produce the following spectrum, allowing us to create the graphic above.

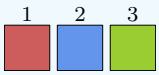
Note.

Using the `Lua` *implementation of* `@prism`*, see the section* ??*, via* `luadraw`*, we can create the palettes below made from the previous one named* `'GeoRainbow'`*. Each instruction used is given below each palette.*

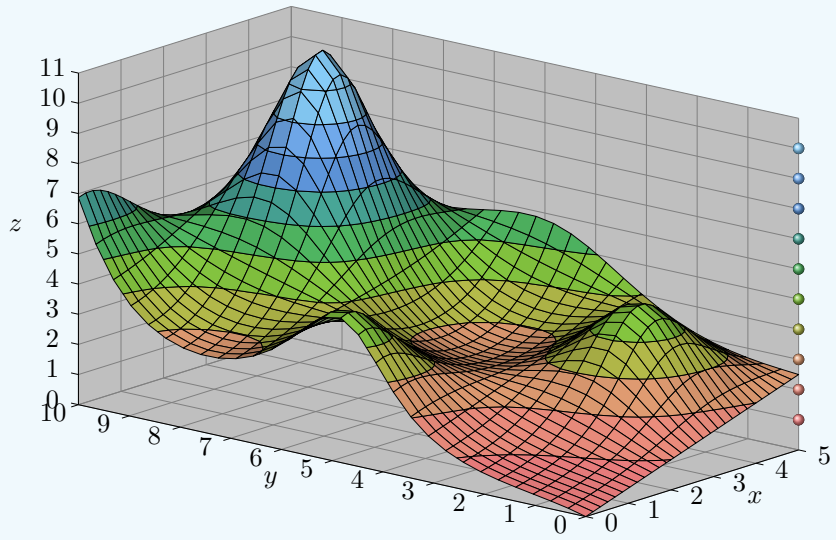



```
palCreateFromPal('GeoRainbow', {reverse = true})
```

```
palCreateFromPal('GeoRainbow', {shift = 3})
```




```
palCreateFromPal('GeoRainbow', {extract = {7, 1, 4}})
```

This features provide remarkable creative flexibility: with the same surface as before, using the setting `palCreateFromPal('GeoRainbow', {extract = {6, 5, 4, 3, 1, 2}})` *instead of*

`palCreateFromPal('GeoRainbow')`, we change the visual tone, shifting from a seaside feel to a snow-covered world.